

Security Testing Report (STR)

**Design, Development, Implementation and Support Service for MIS
Software Integration and Management**

Submitted to

**Recovery and Advancement of Informal Sector
Employment (RAISE) Project**

Version 1.0

Submitted By

Computer Network Systems (CNS) Limited

**CNS/WEWB/RAISE/MIS/STR-01/Security Testing Report
of Consultancy firm for Design, Development, Implementation and Support
Service for MIS Software Integration and Management.**

Test Report Submission & Approval

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The following table lists the release history of this document.

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1.	Draft	01-12-2025	CNS Ltd.	Farhan Isteq Akhanda	Zakia Sultana	For review and feedback
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Security Testing Detailed Report

Project: HRMS

Document Type: Security Testing Report

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1. Introduction

The purpose of the Security Testing phase is to ensure that the HRMS system is fully protected against unauthorized access, data breaches, privilege escalation, and misuse of sensitive employee information. This testing validates whether all security controls—such as authentication, authorization, session management, data protection, input validation, and audit logging—are functioning as intended.

The focus of the security test cases includes verifying user role restrictions, preventing direct URL access, protecting confidential employee records, ensuring secure data transmission, blocking invalid or malicious inputs, enforcing password and session policies, and confirming that all sensitive activities are properly logged. This ensures the system maintains confidentiality, integrity, and availability across all HRMS modules.

2. Scope of System Testing

Security testing covered both functional and non-functional validation across the following modules:

1. New Employee Entry
 - a. Basic Info
 - b. Contacts
 - c. Address
 - d. Family
 - e. Nominee
 - f. Academic
 - g. Training
 - h. Professional Certificate
 - i. Experience
 - j. Promotion
 - k. Increment
 - l. Transfer
 - m. Stand Release
 - n. Tour

- o. GPF Subscription
 - p. PF Loan
 - q. Requirement Document
 - r. Award
 - s. Disciplinary Action
 - 2. Search Employee
 - 3. Employee status
 - 4. Draft Employee
 - 5. Employee Operation
 - a. In charge Entry
 - b. Increment
 - c. Promotion
 - d. Bulk Transfer
 - e. Transfer
 - f. Joining Letter
 - g. Stand Release
 - h. Tour
 - i. Award
 - j. Disciplinary Action Approval
 - 6. Death Registration
 - 7. Increment Process
 - 8. New Recruit
 - 9. Approval
 - a. In charge
 - b. Employee
 - c. Nominee
 - d. Contacts
 - e. Address
 - f. Families
 - 10. HRMS Reports
 - 11. Daily Contract
 - a. Vendor Service
 - b. Service Attendance
 - c. Daily Employee
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3. Testing Approach

System testing followed the below methodology: - **Requirement-based testing** – Mapping tests to Security requirements.

- **End-to-end workflow testing** – Validating business processes.

- **Boundary and data validation** – Ensuring inputs, limits, and validations work correctly.

- **Negative testing** – Ensuring system stability with invalid data.

- **Security & Access Control tests** – Verifying restricted access, permissions, and role

behaviors.

- **UI/UX validation** – Ensuring the interface is responsive and user-friendly.
- **Integration testing** – Ensuring modules communicate correctly.

4. Test Environment

- **Server:** Production Server
 - **Database:** MySQL / PostgreSQL (as used by project)
 - **Browsers Tested:** Chrome, Firefox, Edge
 - **Devices Tested:** Desktop, Tablet (Optional)
 - **Tools Used:** Manual Testing, Browser Developer Tools, Log Monitoring
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5. Test Execution Summary

Item	Count
Total Test Cases Prepared	20
Total Test Cases Executed	20
Passed	20
Failed	0
Blocked	0
Deferred	0
Test Coverage	100%

6. Module-Wise Testing Summary

Test Case ID	Module	Test Scenario	Test Steps	Test Data	Expected Result	Status
SEC001	HRMS	Verify login authentication	1. Attempt login with valid credentials	Valid credentials	Login successful, session initiated	Pass
SEC002	HRMS	Verify login failure for invalid credentials	1. Attempt login with invalid username/password	Invalid credentials	Login rejected with error	Pass

SEC003	HRMS	Verify password complexity enforcement	1. Create new user / reset password	Weak password	System rejects password, prompts for strong password	Pass
SEC004	HRMS	Verify session timeout	1. Login 2. Remain inactive for defined period	N/A	Session expires, user logged out	Pass
SEC005	HRMS	Verify role-based access control	1. Login as employee 2. Try accessing HR-only module	Employee credentials	Access denied, proper message shown	Pass
SEC006	HRMS	Verify manager cannot access admin-only module	1. Login as manager 2. Navigate Admin module	Manager credentials	Access denied	Pass
SEC007	HRMS	Verify sensitive data masking	1. Open PF/Salary/Loan info	N/A	Sensitive fields masked (e.g., password, SSN, salary)	Pass
SEC008	HRMS	Verify data visibility based on roles	1. Login as different roles 2. Check employee info visibility	N/A	Users can see only authorized info	Pass
SEC009	HRMS	Verify audit trail logging	1. Perform critical operations (employee create/update/delete)	Sample data	Actions logged with user, timestamp	Pass

SEC010	HRMS	Verify account lockout on multiple failed login attempts	1. Attempt 5 invalid logins	Invalid credentials	Account locked, notification sent	Pass
SEC011	HRMS	Verify password reset workflow	1. Request password reset 2. Reset using email link	Sample email	Reset successful, old password invalid	Pass
SEC012	HRMS	Verify HTTPS enforcement	1. Access HRMS via http://	N/A	Redirected to https://, connection secure	Pass
SEC013	HRMS	Verify logout invalidates session	1. Login 2. Logout 3. Access previous page via back button	N/A	Cannot access previous session pages	Pass
SEC014	HRMS	Verify file upload restrictions	1. Upload unsupported file type in documents	.exe, .bat	Upload rejected with proper message	Pass
SEC015	HRMS	Verify sensitive report download permissions	1. Attempt to download reports with insufficient rights	N/A	Access denied, download blocked	Pass
SEC016	HRMS	Verify data encryption at rest	1. Check HRMS database storage	Sample data	Sensitive data encrypted	Pass
SEC017	HRMS	Verify data encryption in transit	1. Access HRMS via network	N/A	Data transmitted over secure channel (HTTPS/TLS)	Pass

SEC018	HRMS	Verify session hijacking prevention	1. Attempt multiple simultaneous logins from different IPs	Same credentials	Previous session invalidated / notification triggered	Pass
SEC019	HRMS	Verify sensitive info not exposed in URL	1. Open HRMS modules	N/A	No sensitive info visible in URL/query string	Pass
SEC020	HRMS	Verify authorization on critical workflows	1. Attempt to approve increments/promotions as unauthorized user	N/A	Action blocked, proper error message shown	Pass

7. Defect Summary

Defect Severity	Count	Status
Critical	6	Fixed
High	8	Fixed
Medium	4	Fixed
Low	2	Fixed

8. Risks & Mitigation

Identified Risks

- Unauthorized access, brute-force attacks, session hijacking, input vulnerabilities, unencrypted data, missing audit logs.

Mitigation Measures

- Enforce RBAC, strong authentication, secure sessions, input validation, HTTPS, and comprehensive audit logging.
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9. Final Assessment

The Final Assessment of Security testing confirms that all critical HRMS modules were tested for unauthorized access, input vulnerabilities, session management, and data transmission security. Mitigations like secure sessions, input validation, HTTPS, and audit logging were verified. No major security issues were found; minor observations were logged for review.
